Special Relativity Notes

Pseudo Scalars/Vectors, Dual Tensors and the Metric Tensor

9/10/26

1 Pseudo Vectors: Angular Momentum

Angular momentum: $\vec{L} = \vec{r} \times \vec{p}$

$$L_i = \epsilon_{ijk}\, r_j\, p_k \qquad (i, j, k = 1, 2, 3)$$

$$L_1 = \epsilon_{123}\, r_2 p_3 + \epsilon_{132}\, r_3 p_2$$
$$L_1 = L_x = y\, p_z - z\, p_y$$

with

$$\begin{array}{l|l} r_1 = x & p_1 = p_x \\ r_2 = y & p_2 = p_y \\ r_3 = z & p_3 = p_z \end{array}$$

$$L_2 = L_y = z\, p_x - x\, p_z, \qquad L_3 = L_z = x\, p_y - y\, p_x$$

$$\vec{L} = \vec{r} \times \vec{p} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix}$$

Angular momentum is a **pseudo vector**.

2 Pseudo Scalars: the Scalar Triple Product

$$V^{ijk} = A^i B^j C^k, \qquad V = \epsilon_{ijk}\, V^{ijk} = \epsilon_{ijk}\, A^i B^j C^k$$

Expanding over the six non-vanishing permutations of ϵ_{ijk}:

$$\begin{aligned} V = {} & \epsilon_{123} A^1 B^2 C^3 + \epsilon_{132} A^1 B^3 C^2 + \epsilon_{213} A^2 B^1 C^3 \\ & + \epsilon_{231} A^2 B^3 C^1 + \epsilon_{312} A^3 B^1 C^2 + \epsilon_{321} A^3 B^2 C^1 \end{aligned}$$

Using $\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = 1$ and $\epsilon_{132} = \epsilon_{213} = \epsilon_{321} = -1$:

$$V = A^1 B^2 C^3 - A^1 B^3 C^2 - A^2 B^1 C^3 + A^2 B^3 C^1 + A^3 B^1 C^2 - A^3 B^2 C^1$$

Grouping by A^i:

$$V = A^1(B^2C^3 - B^3C^2) - A^2(B^1C^3 - B^3C^1) + A^3(B^1C^2 - B^2C^1)$$

$$V = \begin{vmatrix} A^1 & A^2 & A^3 \\ B^1 & B^2 & B^3 \\ C^1 & C^2 & C^3 \end{vmatrix} = \vec{A} \cdot (\vec{B} \times \vec{C})$$

The scalar triple product $\vec{A} \cdot (\vec{B} \times \vec{C})$ is a **pseudo scalar**.

3 Helicity

$$H = \vec{\sigma} \cdot \vec{p} \qquad \text{(also a pseudo scalar)}$$

where $\vec{\sigma}$ is the spin and $\vec{p}$ is the momentum.

4 Dual Tensor

$$B_i = \frac{1}{2}\epsilon_{ijk}F^{jk}$$

- B_i — **pseudo vector** (also called the **dual tensor**)
- F^{jk} — 2nd rank, **antisymmetric tensor**

$$F^{jk} = \begin{pmatrix} 0 & F^{12} & F^{13} \\ -F^{21} & 0 & F^{23} \\ -F^{31} & -F^{32} & 0 \end{pmatrix}$$

$$B_1 = \frac{1}{2}\left(F^{23} - F^{32}\right) = \frac{1}{2}\left(F^{23} + F^{23}\right) \qquad (\because F^{32} = -F^{23}) = F^{23}$$

$$B_1 = F^{23}, \qquad B_2 = F^{31}, \qquad B_3 = F^{12}$$

5 Metric Tensor in Minkowski Space

$$g_{\mu\nu} = \begin{pmatrix} \pm 1 & 0 & 0 & 0 \\ 0 & \mp 1 & 0 & 0 \\ 0 & 0 & \mp 1 & 0 \\ 0 & 0 & 0 & \mp 1 \end{pmatrix} \qquad \text{(metric tensor; the two sign conventions)}$$

$$ds^2 = g_{\mu\nu}\, dx^\mu dx^\nu$$

6 General Definition of a Metric Tensor

Q1: In general, how do we define a metric tensor?

Orthogonal basis vectors: $\hat{i}, \hat{j}, \hat{k}$ (or $\hat{e}_x, \hat{e}_y, \hat{e}_z$):

$$\vec{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$$

In general the basis vectors may not be orthogonal, but they have to be **linearly independent**. Let $\vec{\epsilon}_i \rightarrow$ basis vector:

$$\vec{A} = A^1\vec{\epsilon}_1 + A^2\vec{\epsilon}_2 + A^3\vec{\epsilon}_3 = A_1\vec{\epsilon}^1 + A_2\vec{\epsilon}^2 + A_3\vec{\epsilon}^3$$

where $q^i \rightarrow$ generalised coordinates, $\vec{\epsilon}_i \rightarrow$ basis vector:

$$\vec{\epsilon}_i = \frac{\partial x}{\partial q^i}\hat{e}_x + \frac{\partial y}{\partial q^i}\hat{e}_y + \frac{\partial z}{\partial q^i}\hat{e}_z$$

Q2: How do we write the distance element in terms of $\vec{\epsilon}_i$?

Since $\vec{A} = A^1\vec{\epsilon}_1 + A^2\vec{\epsilon}_2 + A^3\vec{\epsilon}_3$ and

$$\vec{\epsilon}_{1,2,3} = \frac{\partial x}{\partial q^{1,2,3}}\hat{e}_x + \frac{\partial y}{\partial q^{1,2,3}}\hat{e}_y + \frac{\partial z}{\partial q^{1,2,3}}\hat{e}_z,$$

an infinitesimal displacement is $d\vec{r} = \sum_i \vec{\epsilon}_i\, dq^i$. The squared distance element is then

$$ds^2 = d\vec{r}\cdot d\vec{r} = \sum_{i,j}(\vec{\epsilon}_i\cdot\vec{\epsilon}_j)\, dq^i dq^j$$

$$\boxed{g_{ij} = \vec{\epsilon}_i\cdot\vec{\epsilon}_j} \qquad \Longrightarrow \qquad ds^2 = g_{ij}\, dq^i dq^j$$

This is the general definition of the metric tensor: the matrix of dot products of the (possibly non-orthogonal) basis vectors — answering both Q1 and Q2 together.

Worked Example: Spherical Coordinates

Completing the example left open in the notes

The notes set up $q^1 = r,\ q^2 = \theta,\ q^3 = \phi$ with

$$x = r\sin\theta\cos\phi, \qquad y = r\sin\theta\sin\phi, \qquad z = r\cos\theta,$$

and ended with "$\vec{\epsilon}_x, \vec{\epsilon}_y, \vec{\epsilon}_z = ?$" unanswered. Below this is carried through to the line element ds^2.

Basis vectors $\vec{\epsilon}_i = \dfrac{\partial x}{\partial q^i}\hat{e}_x + \dfrac{\partial y}{\partial q^i}\hat{e}_y + \dfrac{\partial z}{\partial q^i}\hat{e}_z$:

$$\begin{aligned}
\vec{\epsilon}_1 &= \vec{\epsilon}_r = \sin\theta\cos\phi\,\hat{e}_x + \sin\theta\sin\phi\,\hat{e}_y + \cos\theta\,\hat{e}_z \\
\vec{\epsilon}_2 &= \vec{\epsilon}_\theta = r\cos\theta\cos\phi\,\hat{e}_x + r\cos\theta\sin\phi\,\hat{e}_y - r\sin\theta\,\hat{e}_z \\
\vec{\epsilon}_3 &= \vec{\epsilon}_\phi = -r\sin\theta\sin\phi\,\hat{e}_x + r\sin\theta\cos\phi\,\hat{e}_y
\end{aligned}$$

Metric components $g_{ij} = \vec{\epsilon}_i \cdot \vec{\epsilon}_j$. A direct check shows all cross terms vanish (the coordinate system is orthogonal):

$$\vec{\epsilon}_r \cdot \vec{\epsilon}_\theta = \vec{\epsilon}_r \cdot \vec{\epsilon}_\phi = \vec{\epsilon}_\theta \cdot \vec{\epsilon}_\phi = 0$$

and the diagonal terms are

$$g_{rr} = \vec{\epsilon}_r \cdot \vec{\epsilon}_r = \sin^2\theta + \cos^2\theta = 1$$

$$g_{\theta\theta} = \vec{\epsilon}_\theta \cdot \vec{\epsilon}_\theta = r^2\cos^2\theta + r^2\sin^2\theta = r^2$$

$$g_{\phi\phi} = \vec{\epsilon}_\phi \cdot \vec{\epsilon}_\phi = r^2\sin^2\theta$$

$$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2\sin^2\theta \end{pmatrix} \quad \Longrightarrow \quad \boxed{ds^2 = dr^2 + r^2 d\theta^2 + r^2\sin^2\theta\, d\phi^2}$$

the familiar spherical-coordinate line element.